# Task 01:- Security Operations Center (SOC) Lab – Log Analysis, Monitoring & Detection

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**1. Log Analysis Practice**
**Objective**
To analyze system and application logs to detect suspicious activities such as brute-force attacks using both Windows and SIEM-based approaches.

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A. Windows Event Viewer – Failed Login Detection
Steps Performed

1. Pressed Win + L to lock the system
2. Entered incorrect password multiple times to generate failed login attempts
3. Opened Event Viewer:
    - Pressed Win + R
    - Typed eventvwr and pressed Enter
4. Navigated to:

Event Viewer (Local) → Windows Logs → Security

5. Applied filter:
    - Clicked Filter Current Log
    - Entered Event ID:

4625

6. Clicked OK

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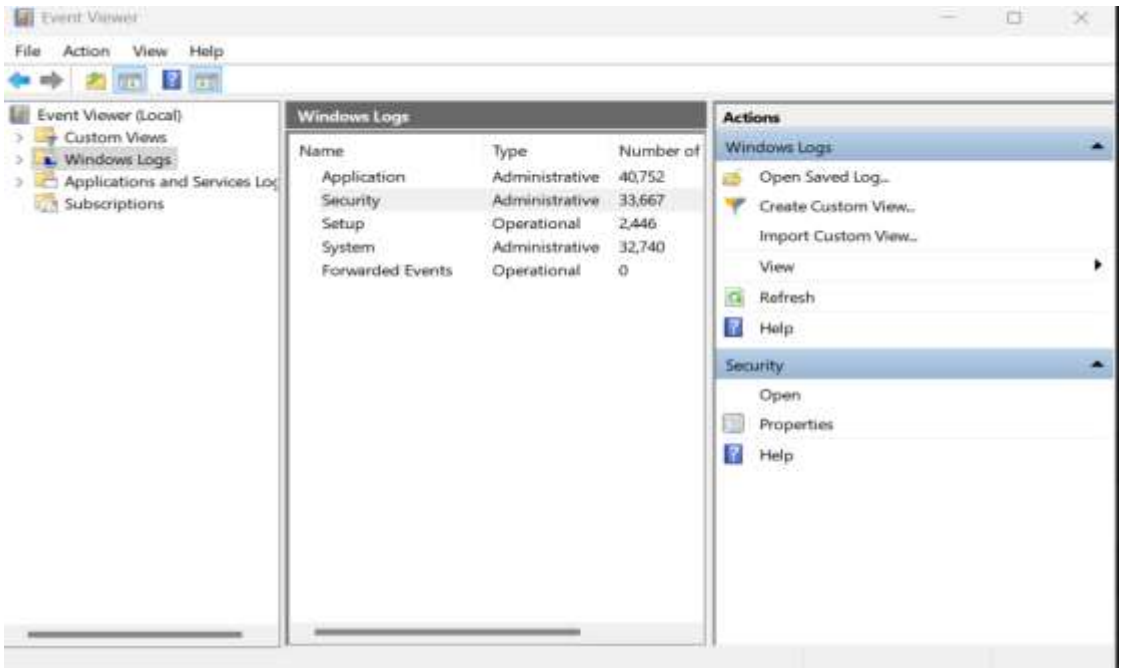
Analysis

- Multiple Event ID 4625 entries observed
- Same user account targeted repeatedly
- High frequency of failed login attempts
- Indicates brute-force attack behavior

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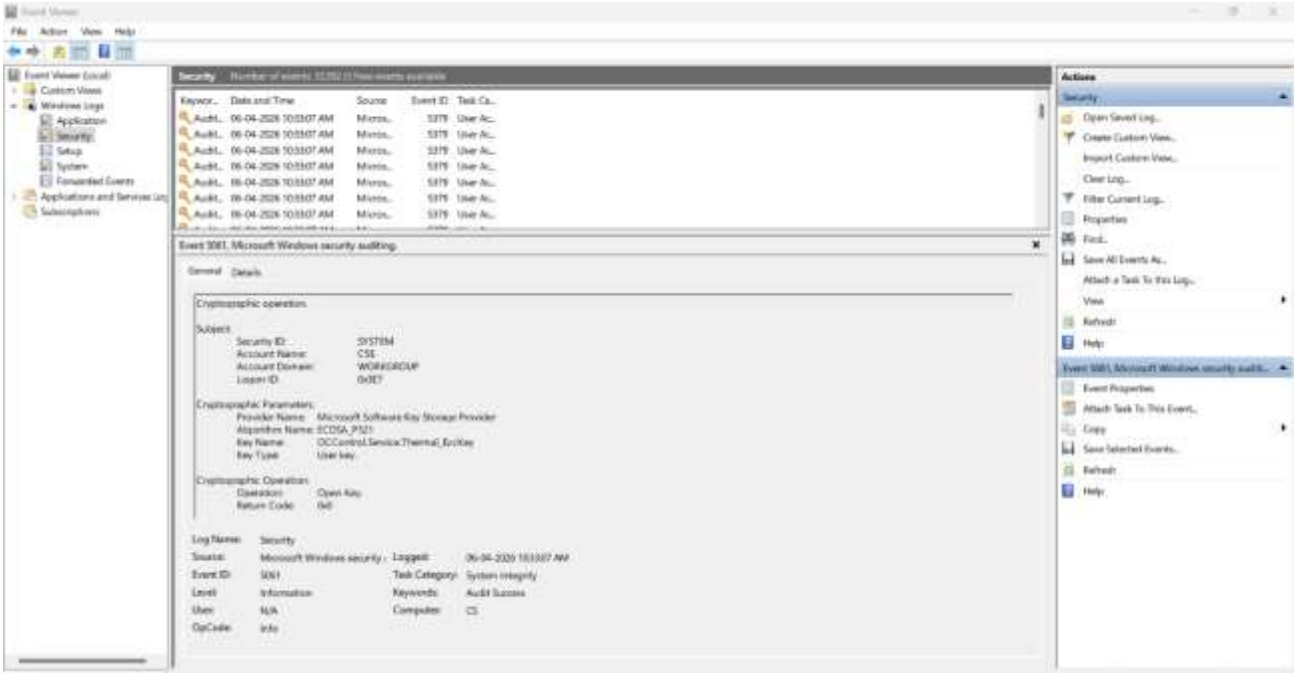

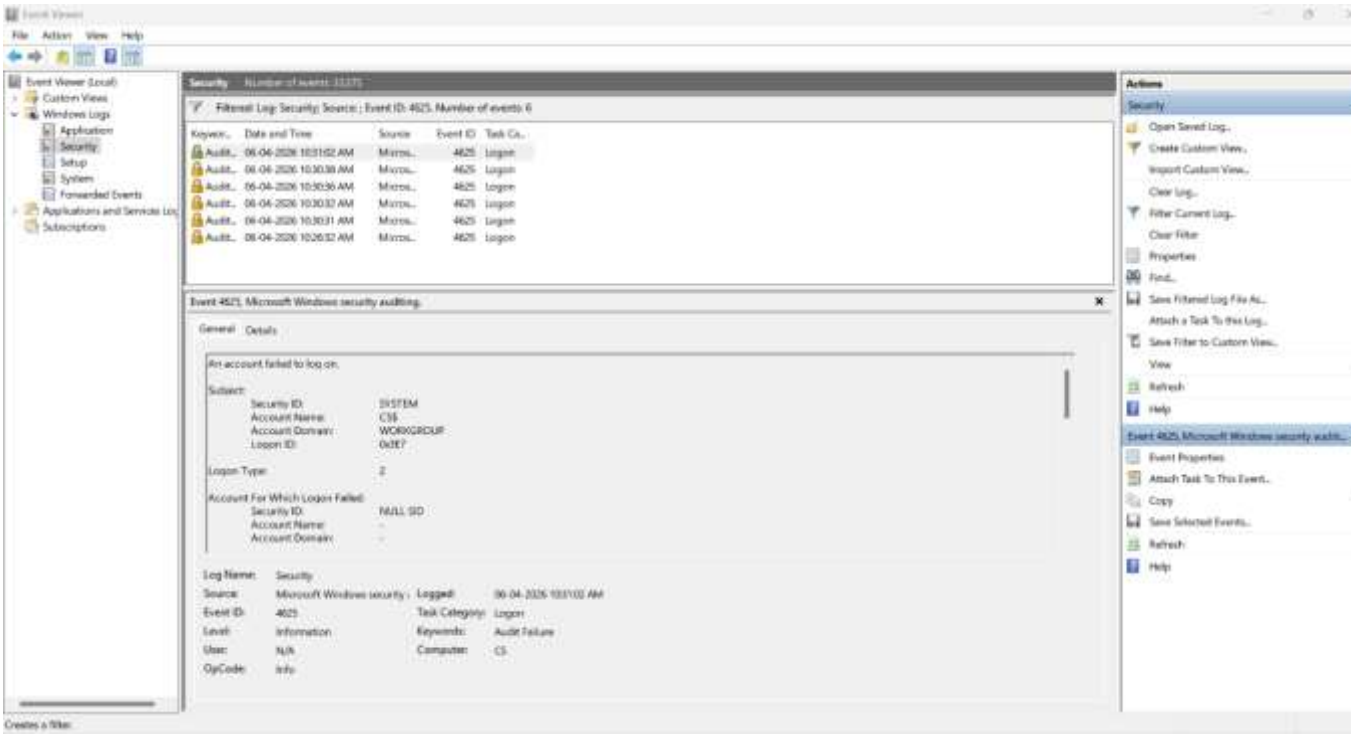

Advanced (CLI Log Extraction)
Executed command in Administrator CMD:
wevtutil qe Security /q:"*[System[(EventID=4625)]]" /f:text

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Event[5]
  Log Name: Security
  Source: Microsoft-Windows-Security-Auditing
  Date: 2026-04-06T10:31:02.9380000Z
  Event ID: 4625
  Task: Logon
  Level: Information
  Opcode: Info
  Keyword: Audit Failure,
  User: N/A
  User Name: N/A
  Computer: CS
  Description:
An account failed to log on.
```

Event ID 7045 (Service Creation)
Filtered logs using:
7045

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**B. Log Analysis Using Kali Linux + ELK Stack**
**Steps Performed**

1. Simulated brute-force attack:

ssh socuser@localhost

- Entered incorrect password multiple times

2. Verified logs:

sudo grep "Failed password" /var/log/auth.log

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Analysis

- Multiple failed SSH login attempts detected
- Same user (socuser) targeted
- Source IP (::1) repeated
- Confirms brute-force attack

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Kibana Log Analysis
Query used:
message: "Failed password"

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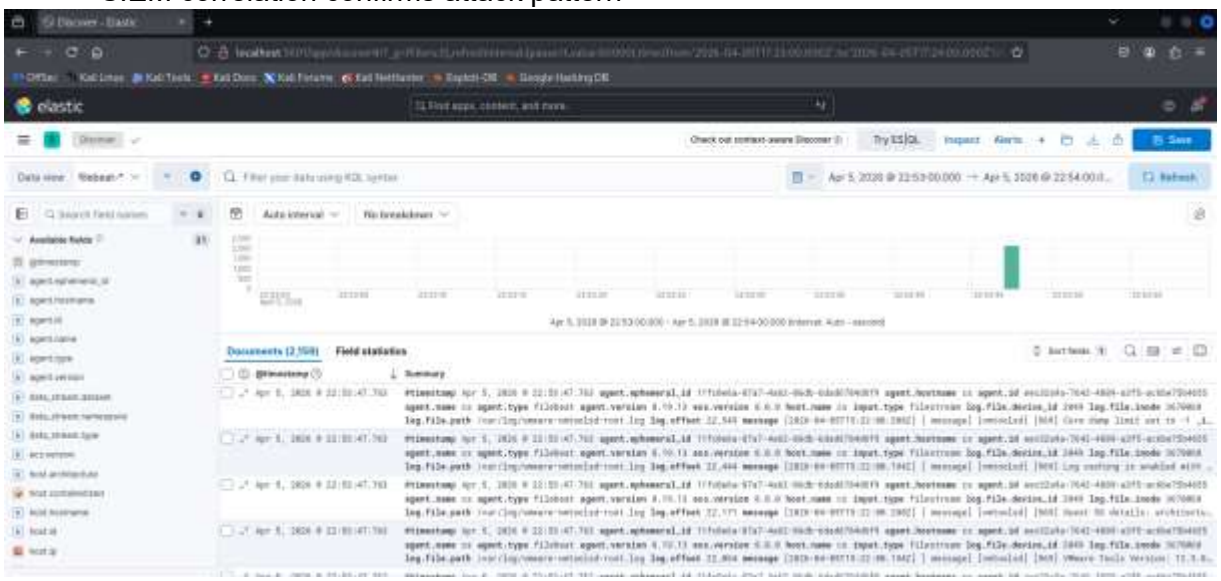
Analysis

- Logs successfully ingested using Filebeat
- Multiple failed login events visible
- SIEM correlation confirms attack pattern

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**C. Browser History Analysis (Zimmerman Tools)**
Steps Performed

1. Downloaded LECmd (Eric Zimmerman Tool)
2. Located Chrome history file:

C:\Users\<user>\AppData\Local\Google\Chrome\User Data\Default\History

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3. Executed:

LECmd.exe -f History

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Analysis

- Extracted browsing history
- Identified visited URLs
- Checked for suspicious URL:

http://test.com

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**2. Document Security Events**

**Objective**

To document detected security incidents in a structured SOC format.

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Template Used

Date/Time Source IP Event ID Description Action Taken

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Recorded Incident

| Date/Time | Source IP | Event ID | Description | Action Taken |
|---|---|---|---|---|
| 2026-04-05 | ::1 | 4625 / SSH logs | Multiple failed login attempts | Monitoring and mitigation suggested |

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Analysis

- Incident categorized as brute-force attack
- Documented for SOC investigation and response

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**3. Set Up Monitoring Dashboards**

**Objective**

To visualize security events and identify attack trends.

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Steps Performed in Kibana

1. Opened Kibana
2. Navigated to:

Visualize → Create Visualization

3. Created:

Visualization 1:

- Top source IPs generating failed logins

Visualization 2:

- Frequency of failed login attempts

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Query Used

message: "Failed password"

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Analysis

- Identified high-frequency login attempts
- Visualization aids faster SOC decision-making

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**4. Configure Alert Rules**
**Objective**
To detect brute-force attacks using threshold-based logic.

Rule Configuration
Rule Name:
Brute Force Detection
Condition:
message: "Failed password"
Threshold:
More than 5 attempts within 5 minutes

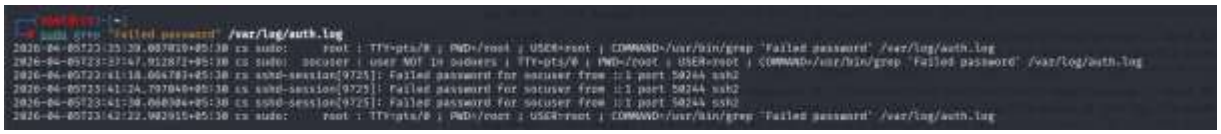
Testing
Executed:
ssh socuser@localhost

- Entered incorrect password multiple times

Analysis

- More than 5 failed attempts detected
- Same user and IP targeted
- Confirms brute-force attack pattern

Alert Validation
Due to lab limitations, alert was validated manually by observing logs exceeding threshold conditions.



Advanced Task
Brute-Force Detection

- Generated failed login attempts
- Verified logs using SIEM and system logs

Zimmerman Tools Practice

- Parsed Chrome history
- Identified browsing activity

Wazuh (Conceptual Implementation)
Rule:
Detect 3 failed logins within 2 minutes
Expected Output:

- Alert triggered in Wazuh dashboard

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**Learning Approach**
**Tools Used**

- Elastic SIEM (ELK Stack)
- Filebeat
- Windows Event Viewer
- SSH
- Zimmerman Tools

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MITRE ATT&CK Mapping

| Technique | Description |
|---|---|
| T1110 | Brute Force |

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Explanation

- Failed login attempts mapped to Credential Access
- Helps identify attacker tactics and behavior

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**Conclusion**
This practical implementation successfully demonstrated:

- Log analysis using Windows and Linux systems
- Simulation of brute-force attacks
- Detection using SIEM queries
- Visualization of attack patterns
- Incident documentation and response

The lab replicates real-world SOC analyst workflow.